

Physicalized Model Weights - cycles 38-38

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Physicalized Model Weights - cycles 38-38

Abstract

Cycle 38 did not open a research cycle. It recorded an `ADMISSION_BLOCKED_NO_CYCLE` state because no admissible trigger was present.

The supplied audit report issued `PIVOT`. This means the campaign boundary remains intact: validating a no-op as research progress would violate the no-null-cycle rule. The controlling endpoint remains `M-PUBLICBASE-SYNTH-1`.

The current endpoint state remains unchanged:

Endpoint field	Current value
<code>current_superiority_claim_count</code>	0
<code>actual_reopen_candidate_count</code>	0
<code>new_reopen_gate_count</code>	0
<code>current_artifacts_reopen</code>	false

Introduction

The physicalized model weights campaign remains closed under current evidence. Earlier validated work found that public MLPerf v6.0 evidence is useful as programmable-baseline prior context, but not as measured hybrid evidence [10]-[14].

Cycle 38 was an admission-control step. In this context, admission control means deciding whether a new research cycle should be instantiated. A new cycle is admissible only if one of the established triggers appears: measured hybrid production/shadow/canary evidence, changed compiled-Verilator capability relevant to `M-PROTO-1`, a materially new primary public-data mapping scope, or a nonduplicative handoff artifact requirement.

No such trigger appeared.

Approach

The reporter gathered the three supplied cycle sessions, the supplied audit report, the current manifest, reference list, report and figure inventory, ledger tail, and `public_baseline_synthesis_summary.json`.

Role	Session ID	Contents
Researcher	fa753246-c9d7-4f9a-a869-df9732c57b36	Emitted
Worker	eef386e1-2178-453c-87bc-384f167d04a4	ADMISSION_BLOCKED_NO_CYCLE. Reported that nothing was built or run because no worker cycle was instantiated.
Auditor	22637560-f9f6-48ac-9c34-68952456ea56	Issued PIVOT and confirmed no commands, files, validators, artifacts, or ledger events were created.

Findings

The researcher did not issue a research brief beyond `ADMISSION_BLOCKED_NO_CYCLE`.

The worker created no files, no milestones, no ledger events, no handoff artifacts, and no validation runs. The worker reported that producing artifacts or rerunning validation would be duplicative because no admissible trigger existed.

The auditor accepted the admission-control behavior but classified the cycle as `PIVOT`, not validated progress. The rationale was that this is not substantive research work. It preserves the trigger-gated endpoint and prevents duplicate endpoint validation or watch-state reconfirmation from being treated as a completed research milestone.

The audit report also noted that parallel fanout was not used. That was appropriate because there were no independent trigger-backed subproblems requiring separate researcher, worker, and auditor loops.

Discussion

Cycle 38 reinforces the boundary established in cycles 35-37: absent a trigger, the correct campaign behavior is no cycle.

The controlling state remains:

- latest substantive validated milestone: `M-PUBLICBASE-SYNTH-1`;
- public MLPerf v6.0 evidence remains programmable-baseline prior context only;
- no measured hybrid evidence has appeared;
- no actual reopen candidate exists;
- no physicalized superiority claim exists;
- no new reopen gate exists.

This does not change the scientific conclusion. Full fixed frontier-model physicalization remains rejected under current evidence. The remaining campaign value is the bounded architecture, verification, failure-mode, and future-evidence-gating package already built in prior milestones.

Open Questions

No new open questions were created in cycle 38.

The standing admissible triggers remain:

- lifecycle-valid measured production/shadow/canary hybrid evidence;
- changed compiled-Verilator capability relevant to M-PROTO-1;
- materially new primary public-data mapping scope not already covered;
- nonduplicative handoff artifact requirement.

References

[10] MLCommons, “MLPerf Inference: Datacenter benchmark documentation,” MLCommons. <https://docs.mlcommons.org/inference/>

[11] MLCommons, “MLPerf Inference v6.0 Results,” MLCommons, 2026. <https://mlcommons.org/2026/04/mlperf-inference-v6-0-results/>

[12] MLCommons, “MLPerf Inference v5.1 Results,” MLCommons, 2025. <https://mlcommons.org/2025/09/mlperf-inference-v5-1-results/>

[13] MLCommons, “MLPerf Inference Results v6.0,” GitHub, 2026. https://github.com/mlcommons/inference-results_v6.0

[14] NVIDIA, “MLPerf AI Benchmarks,” NVIDIA. <https://www.nvidia.com/en-us/data-center/resources/mlperf-benchmarks/>

Appendix: Implementation Details

Code Organization

No new cycle 38 code, data, documents, figures, HDL files, tests, milestones, or ledger events were created.

MANIFEST.md already records the active state as trigger-gated admission control/no-cycle, so no manifest edit was required for this cycle. The manifest snapshot remains:

Category	Count
Authored research scripts	35
Authored tests	32
Authored HDL/support files	4
Authored research docs and diagram sources	34
Ledger events	89
Plan milestones	33

Test Results

The cycle itself ran no validators, matching the audit report.

During report preparation, the reporter ran `python3 -m long_exposure.tools.promise_check . as` a snapshot check. It exited 0 with events: 89, plan milestones: 33, with only the known orphan report warnings.

File and Figure Inventory

No cycle 38 figure exists. Existing figures belong to earlier milestones.

Existing cycle reports now include reports/cycles/report_cycles_35-37.md and reports/cycles/report_cycles_35-37.pdf. The current report is intended for report_cycles_38-38.

Session References

Source	Session ID
Cycle 38 researcher	fa753246-c9d7-4f9a-a869-df9732c57b36
Cycle 38 worker	eef386e1-2178-453c-87bc-384f167d04a4
Cycle 38 auditor	22637560-f9f6-48ac-9c34-68952456ea56

Cross-Reference Map

Origin	Consuming state	Meaning
Cycle 38 researcher session	Cycle 38 worker session	ADMISSION_BLOCKED_NO_CYCLE prevented worker instantiation.
Cycle 38 worker session	Cycle 38 auditor session	No-op worker output confirmed no artifacts or validation runs were created.
Cycle 38 auditor session	This report	Supplies the PIVOT decision and rationale.
public_baseline_synthesis_summary.json	Cycle 38 endpoint summary	Confirms M-PUBLICBASE-SYNTH-1 remains the controlling endpoint with zero/false reopen counters.
MANIFEST.md	Future turns	Preserves the current artifact inventory and trigger-gated admission-control/no-cycle state.